Hangzhou Hikrobot Technology Co.,Ltd.

Hikrobot PCIe Diagnostic Tool

User Manual

HIKROBOT

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Chapter 1 Legal Information

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About this Manual

The Manual includes instructions for using and managing the Product. Pictures, charts, images and all other information hereinafter are for description and explanation only. The information contained in the Manual is subject to change, without notice, due to firmware updates or other reasons. Please find the latest version of this Manual at the Hikrobot website (***https://en.hikrobotics.com***).
Please use this Manual with the guidance and assistance of professionals trained in supporting the Product.

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Chapter 2 Disclaimer

The manual guides you to establish and configure the Software. To ensure the properness of usage and stability of the Software, refer to the contents below and read the manual carefully before installation and operation.
As GIFs and videos cannot be played in PDF document, you are recommended to read the online document.

2.1 Get Support

You can get support from us in the following ways:
- **Official Website**: visit ***https://en.hikrobotics.com/*** to get other related documents.
- **Email**: global.support@hikrobotics.com.

2.2 Symbol Conventions

The symbols that might be found in this document are defined as follows.

Symbol	Description
Note	Provides additional information to emphasize or supplement important points of the main text.
Caution	Indicates a potentially hazardous situation which, if not avoided, could result in equipment damage, data loss, performance degradation, or unexpected results.
Danger	Indicates a hazardous situation which, if not avoided, will or could result in death or serious injury.

Chapter 3 Release Notes

Key updates of each version are as follows.

V4.4.0

This version was released on August, 2024.
Supports XoFLink frame grabber. See ***Tool Introduction***.

V4.2.0

This version was released on June, 2023.
Provides PCIe diagnostic tool to check the PCIe running status of frame grabbers.

Chapter 4 Tool Introduction

The PCIe diagnostics tool (hereinafter referred to as the "tool") can read the memory value of the frame grabber's debugging status in real time.
The Tool supports the following types of frame grabbers:

- Camera Link frame grabber
- Coax Press frame grabber
- GigE frame grabber
- 10 GigE frame grabber (support some models)
- XoFLink frame grabber

The main interface and the available functions/operations are shown below.

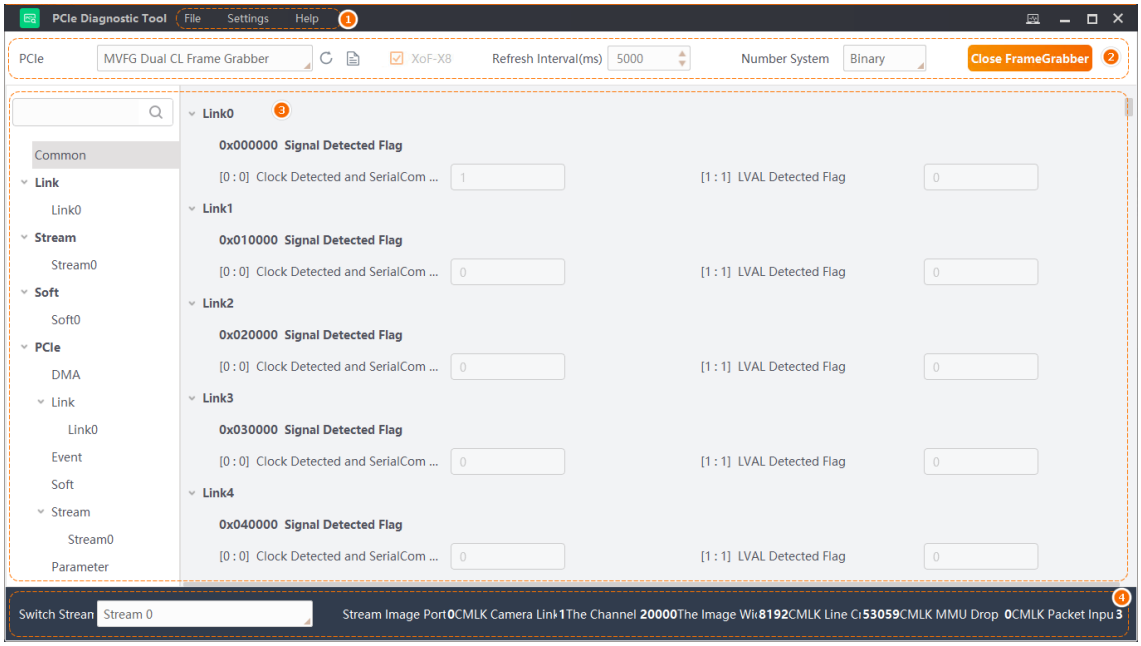

Figure 4-1 Main Interface

Table 4-1 Main Interface Description

No.	Item	Description
1	Menu Bar	The menu bar provides functions of importing/exporting data files, setting general parameters, and other operations for help. ● **File**: You can import external data files or export diagnosis files. ○ Click **File** → **Import** to import external data files to the tool. ○ Click **File** → **Export** to export the diagnosis files to the local PC.

No.	Item	Description
		● **Settings**: You can configure parameters for the diagnosis files to be exported. ○ **Auto Export**: If enabled, the diagnosis files will be exported automatically. ○ **Export Path**: You can configure the saving path for the diagnosis files to be exported. ○ **File Count**: You can configure the number of diagnosis files to be exported automatically. ● **Help**: You can select the language (Chinese or English) for the use of the tool, view the user manual, and view the tool version information.
2	Frame Grabber Settings	You can select a frame grabber for real-time frame grabbing and perform the following operations in the setting area. ● Refresh Frame Grabber: Click to refresh the frame grabber. ● View Frame Grabber: Click to view the frame grabber information. ● Check Memory and CPU: Click in the upper-right corner to check the memory and CPU usage. ● XoF-X8: Enable this when the connected frame grabber is of the XoF-X8 type. ● Set Refresh Interval: You can set the interval for refreshing the debugging status after the frame grabber starts working. The unit is "ms" and the default interval (5,000 ms) is recommended. ● Set Number System: You can set the number system to **Binary**, **Decimal**, or **Hexadecimal** for displaying debugging status.
3	Diagnostic Monitoring	You can monitor the memory value of the frame grabber's PCIe debugging status in real time in the diagnostic monitoring area. See details in ***Diagnostic Monitoring***.
4	Streaming Monitoring	By selecting a stream type from the drop-down list in the **Switch Stream** area, you can monitor the streaming status of the frame grabber in real time, including the stream status and stream transmit status. ● **Stream Transmit Enable**: It refers to the streaming status of the frame grabber and it can be 0 (streaming

No.	Item	Description
		is disabled) or 1 (streaming is enabled). ● **GVSP Packet Input Sof Cnt**: It refers to the number of lost frames of the frame grabber. ● **GVSP Packet Input Data_last Cnt**: It refers to the number of frames transmitted to the tool. **Note** The interface may vary with the frame grabber types. The actual interface shall prevail.

Chapter 5 Diagnostic Monitoring

By connecting frame grabbers with the tool to grab frames, you can monitor the memory value of the frame grabber's PCIe debugging status in real time.
Select a frame grabber model from the drop-down list in the upper-left corner and click **Open FrameGrabber** in the upper-right corner.
You can enter keywords of debugging status or address in the search box on the left and click to search for the target information. The Common page displays the frequent debugging status information.
The tool can monitor 4 types of debugging status for the frame grabber, including **Link**, **Stream**, **Soft**, and **PCIe**.

- **Link**: Transmitted register data.
- **Stream**: Data stream, etc.
- **Soft**: Registers related with Soft software, SDK, serial port configuration, etc.
- **PCIe**: Transmission protocols, APIs, status, counting, etc.

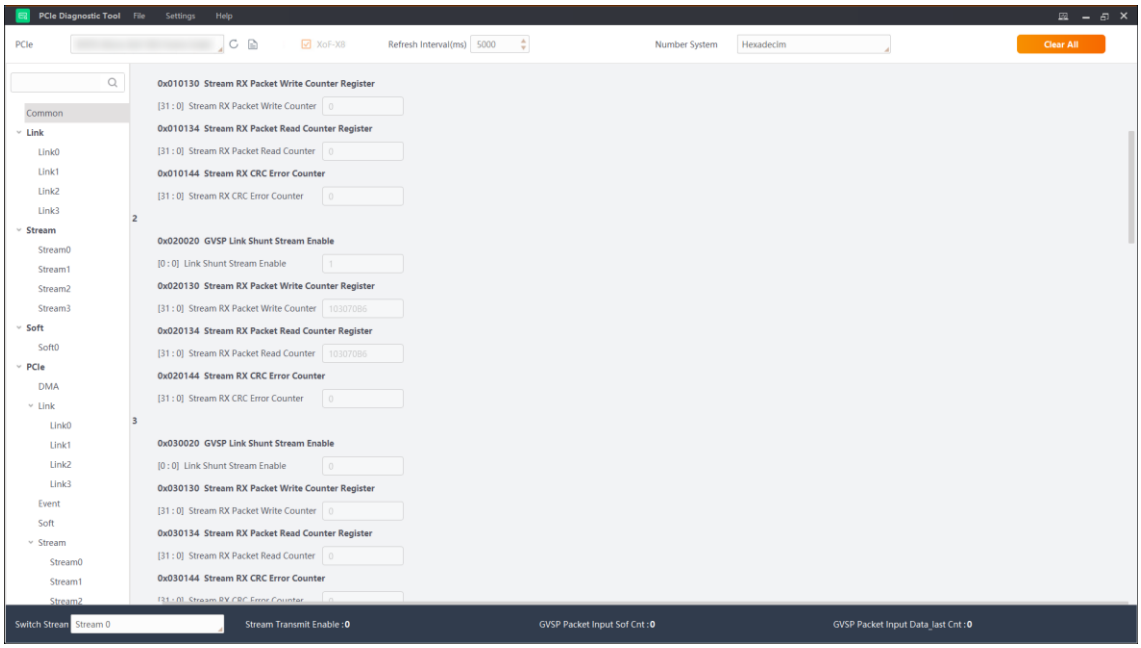

Figure 5-1 Diagnostic Monitoring

Note

Click **Close FrameGrabber** to stop grabbing frames.

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